

The Torus Triple and the Corrections

the calibration stand from first principles

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The «Dynamic Principle» Program — the hodge-laboratory

Standalone theorem monograph · Монография 7/16

1 Setting

The torus is the calibration stand: its triple $(4, 1, 4\pi^2)$ and all derived quantities follow from first principles, with no external premise. The whole correction pipeline is visible at once, and any construction error would show at coarse precision.

Lemma 1 (the torus triple). *The torus supplies the integer triple $(4, 1, 4\pi^2)$: rotation order $n = 4$, carrier index $B = 1$, spectral threshold $\lambda_0 = 4\pi^2$.*

Proof. The rotation $x \mapsto ix$ has order 4: $i^4 = 1, i^2 = -1 \neq 1$. The carrier is the torus itself: $H_2(T^2, \mathbb{Z}) \cong \mathbb{Z}$. The flat Laplacian spectrum is $|m\tau - n|^2$; in the Tate normalization the threshold is $4\pi^2$ (certificate G1). $\square$

Theorem 2 (the corrections derived on the torus). *From $(4, 1, 4\pi^2)$ the pipeline rules give: $\delta = \pi/4 = 0.7853981634$; the rotation holonomy $= i$ (protocol deviation $6.1 \cdot 10^{-17}$ — machine precision); $\delta^2/2 = 0.3084251375$; $\gamma = \delta^4 = 0.3805042619$; $\delta_{\text{eff}} = \delta^5 = 0.2988473484$; $\Delta_{Ch}^{\text{torus}} = 39.4879953935$.*

Proof. Each quantity follows one rule: $\delta = \pi/N$ (the spin double cover); $\delta^2/2$ — the main phase correction; $\gamma = \delta^4/k$ at $k = 1$; $\delta_{\text{eff}} = \delta \cdot \gamma$; the assembly by Δ_{Ch} at $R = 0$. The holonomy: the lift of the rotation permutes the sheets; the eigenvalue on an eigenform is i . All digits reproduce at $dps = 35$ with zero last-digit discrepancy. $\square$

Remark 3 (chessboard equivariance). For the symmetric encoding the μ_4 -equivariance is exact (deviation 0.000 over the 4096×4096 grid); for the $(7, 22)$ encoding it is broken (1.000). The contrast proves the corrections are geometric — they inherit the carrier symmetry.

Protocol data: numbers

The phase family $\delta_T = \pi/N$ reproduced for $N = 2, \dots, 8$; the $\gamma, \delta_{\text{eff}}$ tables for $k \in \{1, 22\}$ built in full; the E6 case corrected: $\Delta = 0.7990$ (not 0.5964); on Klein the same formula gives $\Delta_{Ch} \approx 3.4399$ against the observed 3.443 — deviation 0.0905%.

Summary of results: summary

The torus triple and all derivatives are derived from first principles; the exact torus run calibrates all higher rungs.

Verification: reproduction

`python3 laboratory.py` → “Stands → Torus”; the plot `fig_torus.png`.

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